

Lecture 8

UI review

Outline

- Why review UI?
- Interaction methods
- UI Elements
- UI Principles
- Icon Guidelines

References

- Alan Cooper, About Face 3: The Essentials of Interaction Design

Why review UI?

Why review UI?

- Reduce "Excise" (Unnecessary effort)
- Improve User Efficiency
- Prevent Cognitive Friction
- Ensure Goal-Directed Design

Objectives:

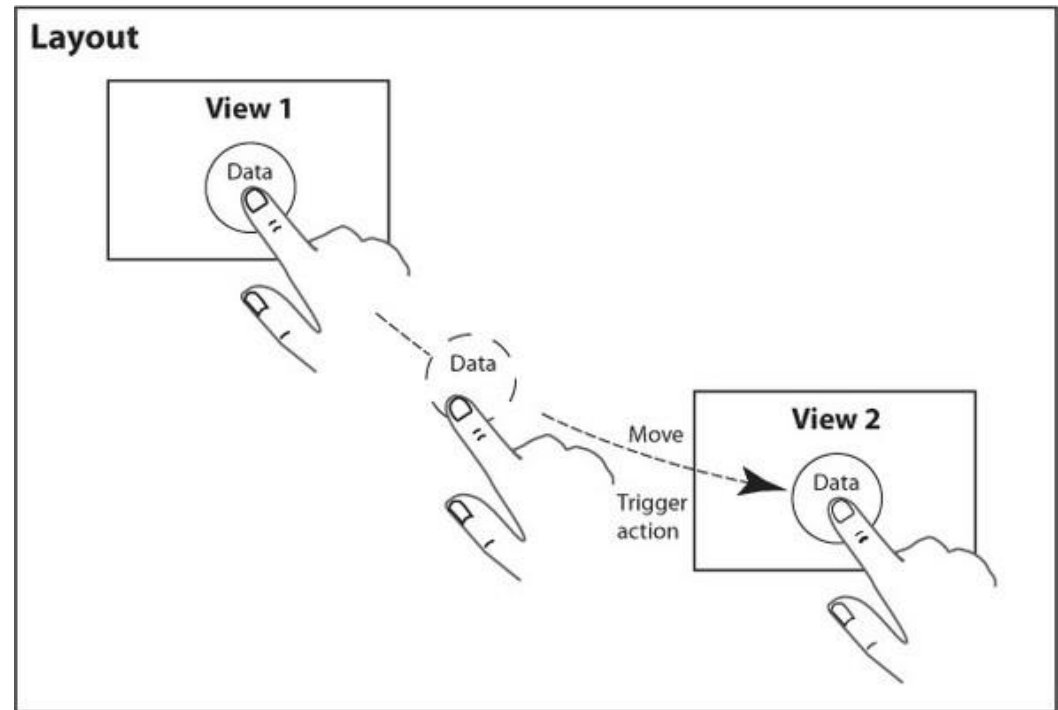
- Evaluate Interaction Methods
- Standardize UI Elements
- Apply Core UI Principles
- Assess Icon Usage & Visuals

Interaction methods

Direct Manipulation

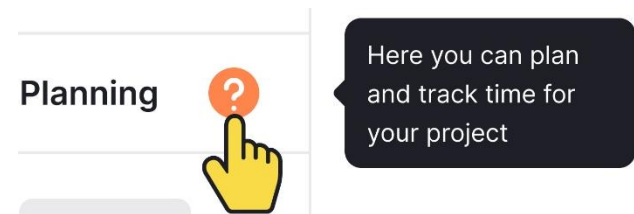
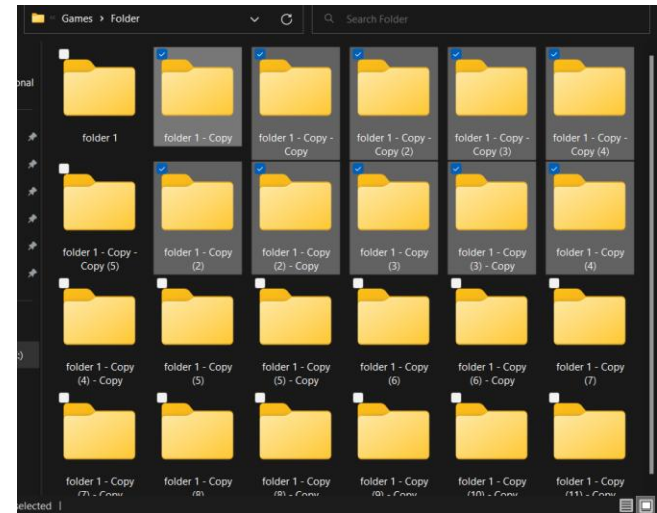
Concept: Users should act directly on objects rather than using complex menus

Example: Dragging a file to a folder vs. using a "Move to" dialog



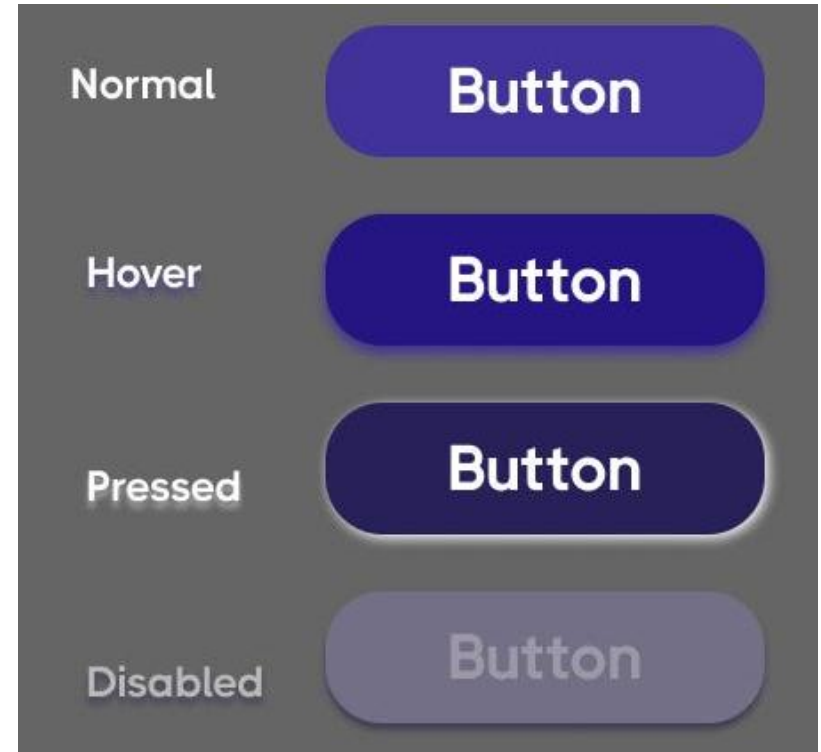
Mouse Interaction & Selection

- **Hover:** Signals "Pliancy" (I am clickable)
- **Single Click:** Selects data or Toggles state
- **Double Click:** Opens objects (Use sparingly on Web)
- **Selection:**
 - *Click:* Select One
 - *Ctrl+Click:* Select Multiple
 - *Shift+Click:* Select Range



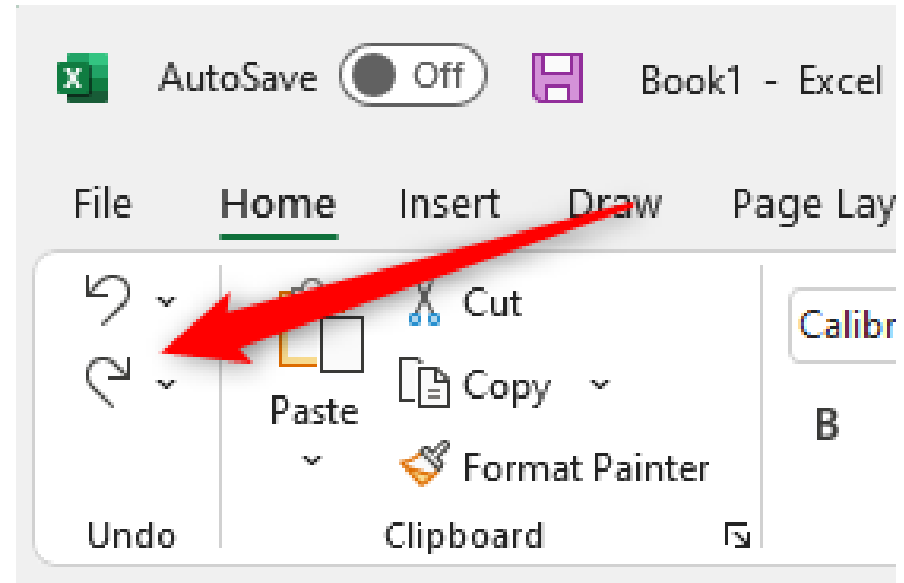
Feedback & Responsiveness

- **Latency Rule:** System must respond in $< 0.1s$
- **Visual Feedback:**
 - Buttons must look "pressed" when clicked
 - Drag actions must show a "ghost" image
- **Goal:** The interface feels solid and responsive



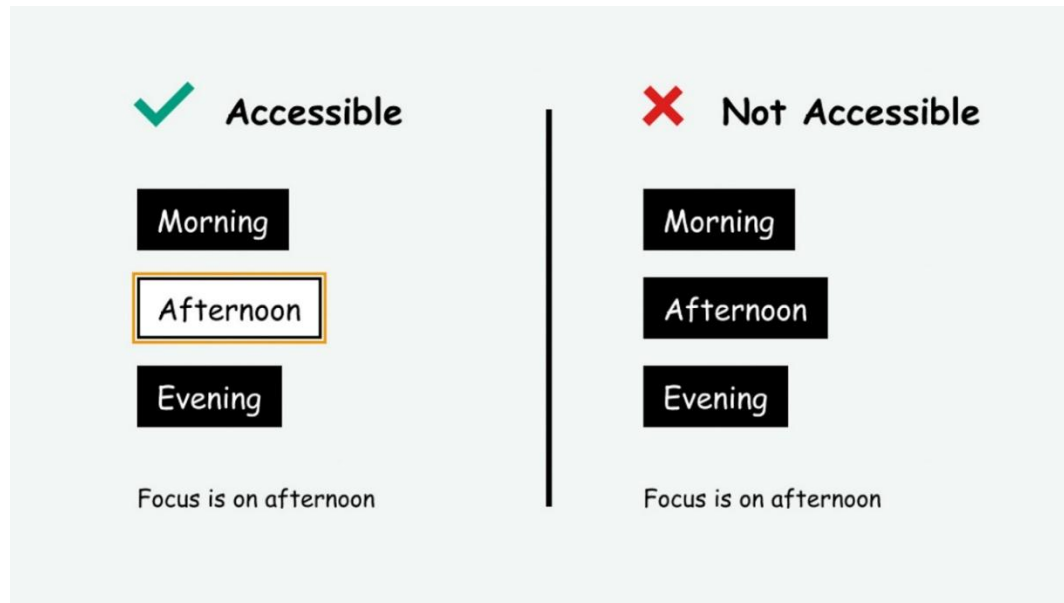
Undo vs. Confirmation

- **The Problem:** "Are you sure?" dialogs interrupt flow
- **The Solution:** Support Infinite Undo
- **Benefit:** Encourages Safe Exploration (Trial & Error)



Keyboard Accessibility

- **Power Users:** Need **Accelerators** (Ctrl+S, Ctrl+C)
- **Navigation:** **Focus Ring** must be visible for Tab navigation
- **Mnemonics:** Underlined letters in menus (e.g., <u>F</u>ile)



UI Elements

Text Inputs

- Text field

- Text area

```
The cat was playing in the  
garden.
```

Buttons



Button with text only



Button with icon only



Button with text & icon

Social buttons



Checkboxes & Radio buttons

- Checkboxes
 - ☐ I have a bike
 - ☐ I have a car
 - ☐ I have a boat
- Radio buttons
 - ☐ Yes
 - ☐ No
 - ☐ Not sure

Dropdown & List boxes

- Dropdown button

Choose a car:

Volvo

Saab

Fiat

Audi

- List box

Today's roster:

Anderberg, Michael

Bouchard, Thomas

Camp, David

Haas, Jonathan

Hanif, Kerim

Jelitto, Jacek

Joseph, Brad

Kohl, Franz

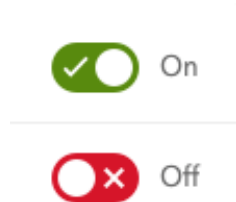
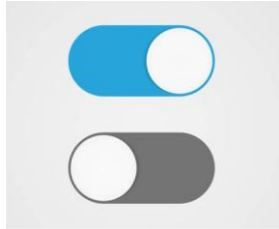
Pai, Jyothi

Raposo, Rui

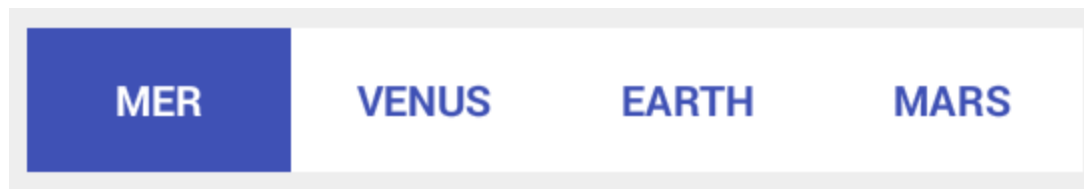
Position: Midfield
Games played: 18
Goals: 4

OK

Toggles



A toggle allows the user to change a setting between two or more states.



Navigational components

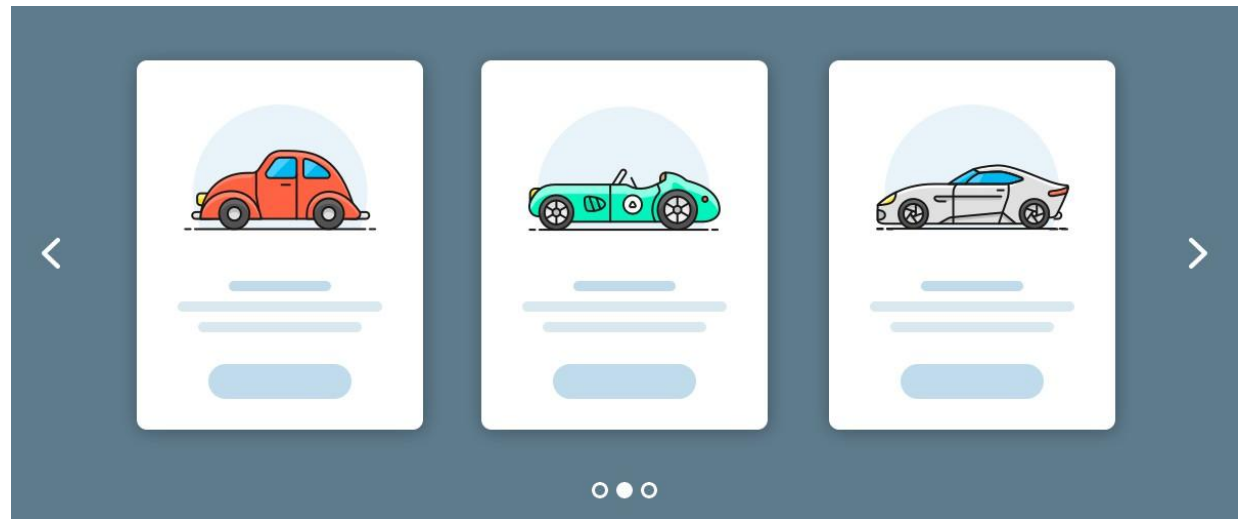
- Breadcrumb

Home ▶ Posts ▶ Category ▶ Article Title

- Pagination

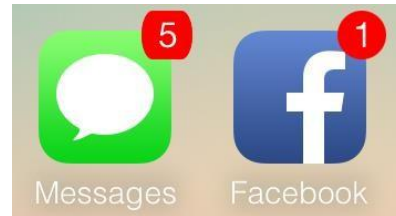


- Carousel

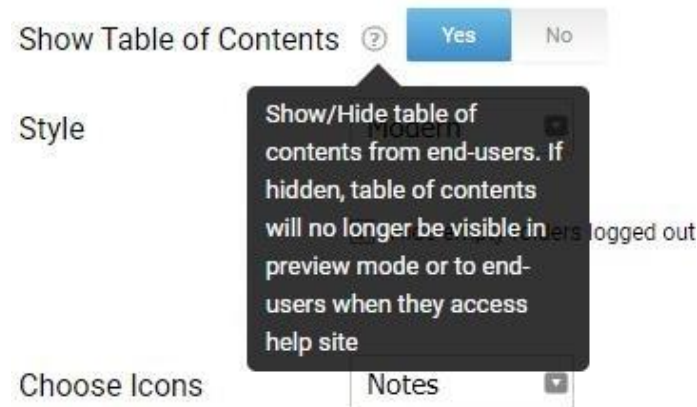


Informational components

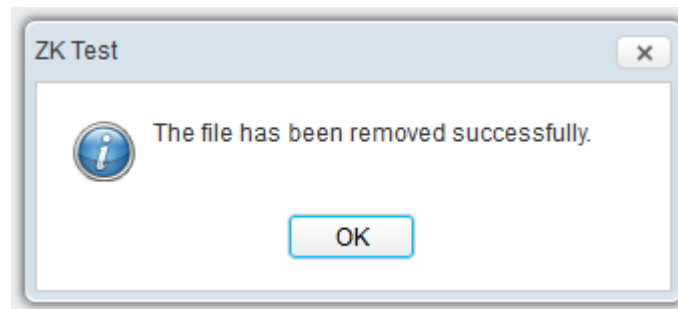
- Badge



- Tooltip



- Message box



UI Principles

UI Principles

Mental Models

- **The Gap:**
 - **Implementation Model:** How the system actually works (Code, Database, Algorithms) — *Complex & Rigid*
 - **User Mental Model:** How the user *imagines* the system works — *Simple & Fluid*
- **The Principle:** The UI should reflect the **User's Mental Model**, not the underlying engineering
- **Goal:** Mask the complexity. Don't force users to understand file systems or database logic

UI Principles

Safe Exploration

- **Behavior:** Users learn by "Trial and Error", not by reading manuals
- **The Principle:** Never punish the user for exploring
- **Requirements:**
 - Support infinite **Undo** (Redo)
 - Avoid confirming obvious actions ("Are you sure?")
 - Provide a "Trash Can" for soft deletes (reversible actions)

UI Principles

Eliminate Excise

- **Definition:** "Excise" is extra work that satisfies the needs of the tool, not the user
- **Examples of Excise:**
 - Resizing or moving windows constantly
 - Dismissing unnecessary pop-up dialogs
 - Re-entering data that was already provided
- **The Principle:** The UI should be transparent.
Minimize navigation and management tasks to **Zero**

UI Principles

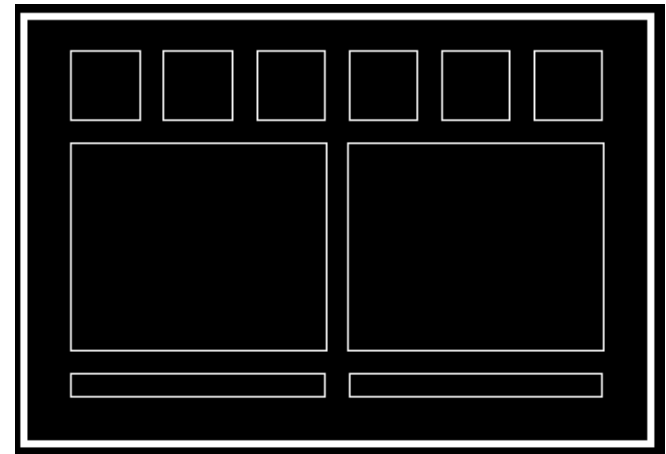
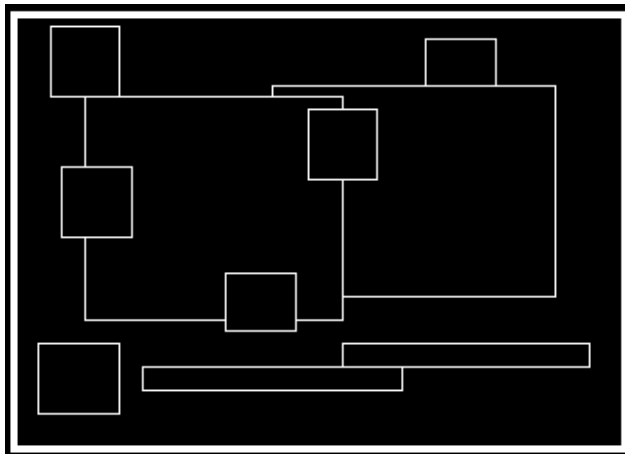
Smart Software (Memory)

- **The Problem:** Most software has “Amnesia”—it forgets everything when closed
- **The Principle:** “Do what I did last time.”
- **Requirements:**
 - Remember window position and size
 - Remember the last opened file/view
 - Remember toggle states and settings
- **Goal:** Treat every restart as a continuation, not a new beginning

UI Principles

Visual Hierarchy

- **Reality:** Users **scan** screens; they do not read them
- **The Principle:** Guide the user's eye using visual weight
 - **Size:** Important elements must be larger
 - **Contrast:** Primary actions must stand out
 - **Position:** Key information should be in the “F-Pattern” or center



UI Principles

Affordance

- **Definition:** Visual properties that indicate how an object is used
- **Examples:**
 - Raised (3D) effect = “**Click me**”
 - Recessed (Inset) effect = “**Fill me**”
 - Grab handles = “**Drag me**”
- **The Principle:** Form follows function. Don't hide functionality behind flat, ambiguous graphics



Flutter Material Button Migration

Flat Button

Raised Button

Outline Button

Text Button

Elevated Button

Outlined Button

UI Principles

Responsiveness

- UI adapt to different devices
- Touch targets usable on mobile
- Elements properly resized

UI Principles

Consistency

- Components consistent
- Naming conventions consistent
- Styled consistently

UI Principles

Visual Design

- Avoid contrast sufficient for readability
- Readable and consistent font
- Intuitive icons

Icon Guidelines

Design Strategy & Metaphors

- **The Role:** Rapid recognition (faster than text) & space saving
- **Metaphors:** Use real-world objects users recognize
 - *Examples: Trash can (Delete), Folder (Directory), Gear (Settings)*
- **Clarity over Beauty:**
 - Readability at 16px/24px is priority
 - Avoid excessive gradients or shadows that blur at small scales.

Usability & Ambiguity

- **Distinctiveness:**
 - Icons must have distinct silhouettes (shapes) to avoid confusion
 - Don't use similar-looking icons for different actions
- **The Golden Rule of Labeling:**
 - **Always** use text labels or tooltips
 - **Reason:** Icons are inherently ambiguous (e.g., Does a "Star" mean Favorite, Rating, or New?)

Icon States & Feedback

- **Visual Feedback is Key:**
 - **Normal:** Standard visibility (high contrast)
 - **Hover:** Glow, brighten, or slight lift (Signals Pliancy)
 - **Active/Pressed:** Darker, inset, or size reduction
 - **Disabled:** Greyscale or 'Ghosted' (Reduced opacity)

Summary

- Why review UI?
- Interaction methods
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